HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and debugged **Linux kernel modules** for endpoint security features, focusing on system calls and process monitoring across multiple distributions.
- Implemented **eBPF programs** to enhance threat prevention capabilities, integrating them with kernel event streams for real-time analysis.
- Designed and maintained **user-space APIs** that interact with kernel-level components, ensuring seamless data exchange for security applications.
- Debugged complex **kernel-level issues** related to memory management and concurrency, identifying root causes for system-level instability.
- Contributed to the **build and packaging** of kernel components for Red Hat and Debian environments, ensuring compatibility and streamlined deployment.
- Collaborated with QA and security teams to validate kernel functionality and address compatibility concerns across x86_64 and ARM64 architectures.
- Wrote **Python scripts** to automate kernel testing processes and analyze performance metrics, reducing manual testing effort by approximately 30%.

SKILLS

Core Technologies – C, C++, Linux Kernel, eBPF, System Calls, Memory Management, Concurrency, Processes, Scheduling

Operating Systems – Linux (Red Hat, Debian, Ubuntu, SUSE)

Architecture – x86_64, ARM64

Development Tools – GDB, Git, Jenkins, Docker

Scripting & Automation – Python, Shell Scripting

System Subsystems – Filesystems, Networking, Kernel Modules

PROJECTS

Linux Kernel Security Module for Endpoint Protection

- Developed a custom **Linux kernel module** in C/C++ to enforce application control policies, integrating with the kernel's LSM framework.
- Implemented event hooks for filesystem operations and process execution, filtering actions based on allowlists and blocklists managed via a configuration file.
- Designed a companion **Python tool** for generating and validating policy configurations, significantly reducing manual errors during policy updates.

eBPF-based Network Traffic Monitoring

- Created **eBPF programs** to capture and analyze network packets at the kernel level, focusing on identifying anomalous communication patterns.
- Developed a user-space application in C++ to process eBPF tracepoints, aggregating traffic statistics and flagging suspicious connections.
- Validated the monitoring solution against known network attack vectors, achieving a detection rate of approximately 85% for simulated intrusion attempts.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)